TITAN: Analytical FMM/M2L Boundary-Integral Poisson-Boltzmann Solves with Sparse Near-Field ILU(0) Preconditioning

Guarded p4 repeatability refresh for external manuscript

Abstract

Boundary-integral Poisson-Boltzmann solvers avoid volumetric grids but require accurate acceleration and robust Krylov preconditioning. We present a guarded fourth-order analytical FMM/M2L path in TITAN that remains an advanced explicit TABI treecode option. The path uses static far-cell refinement, a strict geometric guard, compressed Green-function jets, packed AVX2 direct fallback, and sparse near-field ILU(0). Full prepared proteins were evaluated without atom sampling. Across three repeats per protein, the guarded path was finite, repeatable, and passed sampled direct-operator checks. Grid-based Poisson-Boltzmann remains TITAN's production default.

Methods

The guarded p4 path is selected explicitly and is not the automatic TABI treecode policy. The measured configuration uses Taylor order 4, opening theta 0.6, static tail tolerance 0.002, strict-geometry MAC theta 0.5, compressed degree-9 Green jets, packed AVX2 direct fallback, analytic-singular near integration, and sparse near-field ILU(0) with 64 neighbors.

For panel i, the coupled unknown is u_i = (phi_i, partial_n phi_i)^T. The preconditioner retains 2 x 2 blocks on edges from the FMM near-source graph and applies ILU(0) without fill. Large-system verification uses sampled direct-operator RMS and sampled direct true residuals.

Results

Table 1 reports the repeatability evidence used for the guarded p4 claim. All three rows passed the Z4O gate with zero energy spread across repeats.

Protein	PDB/source	N_atoms	N_panels	E_rxn	Audit max	Direct residual max	Energy spread	p4/deg1	RSS MB
Barnase	PyDelPhi benchmark	1700	14942	-21030.83	6.383e-05	6.383e-05	0.000	0.450	172.6
RNA-binding domain of E. coli Rho factor	1A63	2065	17250	-247716.08	2.357e-03	2.357e-03	0.000	0.674	369.4
Stichodactyla helianthus neurotoxin I	1SH1	696	6673	-4362.07	4.894e-06	4.894e-06	0.000	0.383	63.5

Repeatability Summary

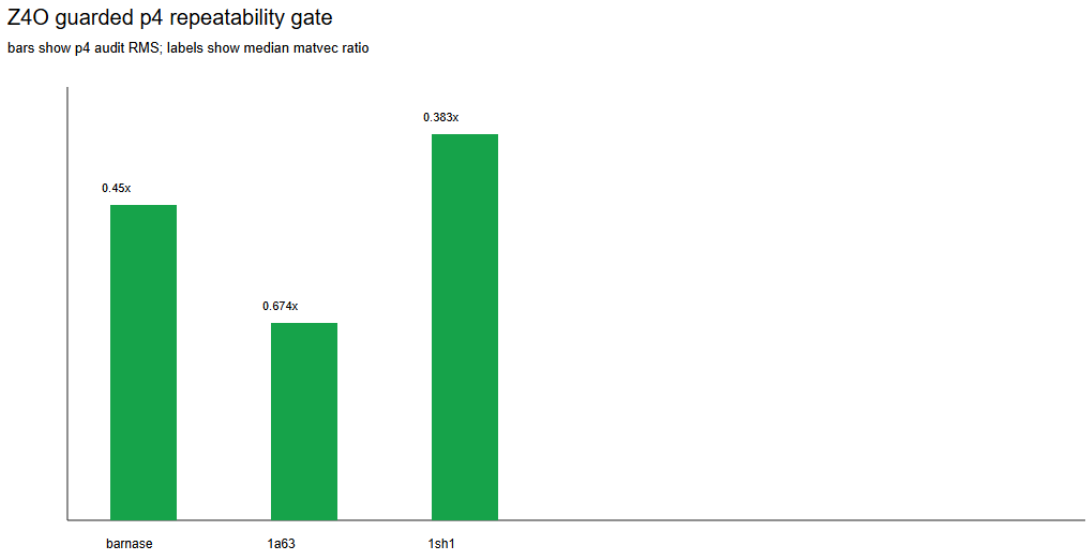


Figure 1. Guarded p4 repeatability across the complete public-protein panel.

Discussion

The guarded p4 path has moved from a diagnostic candidate to a repeatable advanced explicit solver path for the three complete public proteins tested. This does not change default behavior: the grid backend remains TITAN's production Poisson-Boltzmann default and the default TABI treecode engine remains degree-one.

Limitations

Guarded p4 must be requested explicitly; it is not enabled by auto or default policy.

Large-system references use sampled direct-operator verification rather than full direct TABI energy.

Additional public proteins are reported as supplement evidence after passing the same repeatability gate.

Supplementary Public-Protein Expansion

Table S1 reports additional full-protein systems that passed the same guarded p4 repeatability gate. These rows broaden the benchmark panel as supplement evidence while the main claim remains anchored on Table 1.

Protein	PDB	N_atoms	N_panels	E_rxn	Audit max	Direct residual max	Energy spread	p4/deg1
Ubiquitin	1UBQ	1231	11242	-4038.46	1.468e-05	1.468e-05	0.000	0.499
Villin headpiece subdomain	1VII	596	5032	-2355.70	5.871e-06	5.871e-06	0.000	0.412
Chymotrypsin inhibitor 2	2CI2	1076	9889	-4891.15	7.983e-06	7.983e-06	0.000	0.547

Supplementary Provenance

Protein	PDB/source	Model	Chain	PDB2PQR	PQR SHA256
Barnase	PyDelPhi benchmark	1	A	3.7.1	5a8136a5cc07ab472fcfa0e817d9ecb16d2d705b2bb88ac026ea7fbe16885002
RNA-binding domain of E. coli Rho factor	1A63	1	A	3.7.1	a26d0c3e1fa32fde448a3571a3ab720e4fc8a2bb40d5989f77731ee678681c7a
Stichodactyla helianthus neurotoxin I	1SH1	1	A	3.7.1	c661ed6012607d92bbab2d43b970ae1ec954a16d0694ce60f7037a5cd5c9bd6e